MIL-S-20603
18 January 1952

MILITARY SPECIFICATION

SHOT, TARGET PRACTICE, 37 MM.,
M51, METAL PARTS ASSEMBLY

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army PXS-701 (Rev. 3)
30 September 1941

This specification consists of this cover sheet and Specification PXS-701 (Rev. 3), dated 30 September 1941, together with Amendment-1, dated 2 May 1942, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
Army-O

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Picatinny Arsenal
Tentative Specification

PXS-701 (Rev. 3)
September 30, 1941
Amendment-1
May 2, 1942

SHOT, TARGET PRACTICE, 37 MM., M51, METAL PARTS ASSEMBLY

The following changes should be made in Specification PXS-701 (Rev. 3) dated September 30, 1941:

Page 2, Paragraph F-1, delete and substitute the following:
"F-1. Inspection.-
F-1a. Inspection shall be as prescribed in U.S. Army Specification, No. 50-0-1, General Specification for Ammunition Except Small Arms Ammunition, and Federal Specification No. QQ-M-151, Metals; General Specification for Inspection of.
F-1b. The inspector shall assure himself that the rotating band of each shot has not been damaged by being nicked or burred."

Page 3, Paragraph G-1a, add, "Lot number of contents shall be marked plainly on each container."

Page 3, Paragraph G-1b, delete, and substitute the following:
"G-1b. Data cards.-Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight shall be prepared for each lot of accepted shot in accordance with the form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket, and six cards completely filled out shall be furnished the inspector.

DATA CARD

Kind: Shot, Target Practice, 37 mm., M51, Metal Parts Assembly. Ammunition Lot No._____
Quantity in Lot_____
Drawing No._____ Rev._____ Spec. No._____ Rev._____
Manufactured by_____ Date_____
Contract No._____ Order No._____
Packed_____ in a box.
Remarks:_____

Certified to by_____, Inspector,
Print here name of inspector, district, arsenal, or loading plant, and sub-district where applicable."

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Picatinny Arsenal
Tentative Specification

PXS-701 (Rev. 3)
September 30, 1941.

SHOT, TARGET PRACTICE, 37 MM., M51, METAL PARTS ASSEMBLY

A. APPLICABLE SPECIFICATIONS.

A-1. Ordnance Department drawing, No. 75-2-280, and all drawings and specifications listed or referred to thereon form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPE.

B-1. This specification covers one type.

C. MATERIAL AND WORKMANSHIP

C-1. Materials and parts shall be in accordance with applicable specifications and drawings.

C-2. The interior of the tracer cavity shall be smooth and clean. When cleaning is necessary, the method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

C-3. All shot bodies and windshields shall be sound and free from cracks and other defects.

C-4. The rough rotating band shall be in the form of a solid ring, annealed, and carefully pressed into the band seat so as to completely fill it. The band shall be applied cold.

C-5. The machined rotating band shall be carefull protected from nicks, burrs, and other similar damage.

C-6. The painting and marking shall be in accordance with the drawings.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. The shot shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Rotating bands.- The rotating bands shall be properly seated.

E-3. The material of the rotating band shall show in test specimens taken from bands removed from the shot, the following physical properties:

Tensile Strength 1 lb. per sq. in., min.	Percent Elongation Min. in 1 inch	Bend Test Degrees
35,000	15	180

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection.- Inspection shall be as prescribed in U. S. Army Specification, No. 50-0-1, General Specification for Ammunition Except Small Arms Ammunition, and Federal Specification No. QQ-M-151, Metals; General Specification for Inspection of.

F-2. Size of lots.-Twenty thousand maximum. A smaller quantity may constitue a lot at the discretion of the inspector. The first lot manufactured on any contract shall not exceed 2,000 shot, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

F-3. Rotating band tests.- The rotating bands shall be tested as follows:-

F-3a. The inspector shall assure himself that the rotating bands are properly seated. Before finish-machining, tap the band of each shot with a 3-ounce steel hammer at several points and remove one band from approximately every 2,000th shot for examination of the seating. In the tapping test, a hollow sound accompanied by an unusual rebound of the hammer indicates an improperly seated band. All shot, where the test indicates improperly seated bands, together with the shot from which the bands were removed for examination of the seating, may be rebanded and re-submitted for test.

F-3b. One rough band (before machining) shall be removed from each 2000 shot and subjected to the physical tests prescribed in Par. E-3. Bands removed for the seating test may be used. The band shall be carefully removed from the shot and carefully flattened without hammering. The test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to hold to the minimum any additional cold working. Test specimens shall be machined to the form and dimensions prescribed by Specification No. 50-27-1, Blanks, Rotating Band, For Projectiles.

F-3c. For the bend test, each band or section thereof, of full width and thickness, shall be bent cold by pressure, without hammering, in the direction of curvature of the band, until it tightly encloses 180° of a rod equal in diameter to the thickness of the band.

F-3d. If any specimen fails to meet the tensile or elongation requirements in the physical test, or if any band cracks, in the bend test, a retest may be made at the request of the contractor. Double the number of samples shall be used for the retest and if failure occurs in the retest, the lot of shot shall be rebanded and resubmitted for test. The retest will be confined to the same phase of testing in which failure occurred in the original test.

F-4. Proving ground test.- Twenty-five sample shot shall be selected by the inspector from each lot and tested as follows:

F-4a. The sample shot shall be loaded to service weight with an inert filler, and fired through a cardboard or plywood screen placed approximately twenty-five feet in front of the muzzle. The shot shall be fired from a gun for which the shot is standard, using a propellent charge that will give 12 percent excess chamber pressure over the rated maximum pressure for the gun.

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F-4b. Examine the cardboard or plywood screen to determine whether the rotating band is remaining in position and the shot does not break up.

F-4c. The proving ground test specified in preceding paragraphs is not a condition for acceptance of a lot of shot, but the results of this test will be carefully scrutinized by the Chief of Ordnance as being indicative of the methods of manufacture employed by the contractor as governed by the provisions of section C.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packing.-

G-1a. The shot shall be packed in commercial type containers so constructed as to insure acceptance by common or other carrier, for safe transportation at the lowest rate to the point of delivery. Each shot shall be protected from coming in contact with another shot by means of suitable partitions. The packing container shall be sufficiently rigid to withstand transportation and permit storing in tiers at least 10 feet high without damage to the containers or contents. The gross weight of any container shall not exceed 200 lbs.

G-1b. Data card.- One data card, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, and properly filled out according to the form below (except quantity in lot) shall be placed in each packing box. Two of these cards completely filled out for each lot shall accompany each shipping ticket, and six cards completely filled out shall be furnished the inspector.

DATA CARD

Kind: Shot, Target Practice, 37 mm., M51, Metal Parts Assembly Ammunition Lot No._____
Quantity in Lot_____
Drawing No._____ Rev._____ Spec. No._____ Rev._____
Manufactured by_____ Date_____
Contract No._____ Order No._____
Packed_____ in a box.
Remarks:_____

Certified to by_____
Inspector

(Note.- When the contractor for metal parts for the M51 Shot is also the contractor for loading, assembling, and packing of the shot, no data cards will be required for the metal parts).

G-2. Marking.- The marking for shipment shall be in accordance with U. S. Army Standard Specifications for Marking Shipments, No. 100-2.

H. NOTES.

H-1. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Note.- Copies of this specification may be obtained from the Office, Chief of Ordnance, Washington, D.C.

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